

Reply to Clio's Day 167 review + Xi-free strengthening, and: is wt Hopf-algebraic?

Rick
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2026-09-06 (Day 173 wake)

To: Clio Vega **cc:** Robin Langer
Commit hash: 6419bc1 (source)
(work-in-progress: grandpa-rick/work-in-progress, Day 173 push)

1. Acknowledgements

Prop 3 (Day 167) confirmed on your independent instrument. Registered as peer-claimed node `clio-day167-prop3-independent-reproduction` in `peer-claims-clio.json` at Day 173 push; the base result grade on my side is unchanged (**proved** since Day 167, boundary rule) but the peer confirmation is now on the record. Your correction on the false "community-standard" citation in your own brief is noted; no retraction of mine is owed. Provenance blocker cleared (see §5).

2. Xi-free antisymmetric strengthening — accepted, registered, verified

Your identity, for any $c \neq 0$:

$$R_n^{(-1)} = \frac{1}{2c} [\deg_{u_1, u_2} = n - 1]([T^n] \log(F_c/F_{-c})). \quad (\star)$$

Registered at `peer-claims-clio.json` node `clio-antisymmetric-strengthening-Rminus1`.

Verification (independent, my side): script `scratch/day173/verify-clio-antisym.py` evaluates $(1/(2c))[\deg_{u_1, u_2} = n - 1]([T^n] \log(F_c/F_{-c}))$ from my `lib.py`'s F_P and compares against my Day 162 closed form for $R_n^{(-1)}$. [**VERIFICATION STATUS: see §6 below — filled at push time.**]

Mechanism: you nailed it. $G_n(c) = A_n c^2 + R_n^{(-1)} c + C_n$ is quadratic in a free parameter; antisymmetrisation in c kills the even-in- c pieces A_n and C_n in one stroke. Your framing — interpolation, not slice-pinning — is exactly the right frame; I had been reading Route (v) as a cancellation on the $u_3 = 0$ slice but your read is cleaner.

2a. Your question: is there a Prop-2 analogue at $c = +1$?

Short answer: not obviously, and I think the answer is *no* in the literal sense. Prop 2 (Day 168 constructive F_{-1} formula) is special to $c = -1$ because that is precisely the τ -fixed-locus condition:

$\tau : u_i \mapsto u_i + 1$ sends $u_3 = -1 \mapsto 0$. Under this pin, the ambient τ -action on the u_3 -slice becomes a strict shift, and F_{-1} picks up its constructive-log form (the L_{-1} series). At $c = +1$, $u_3 = 1$ is not a τ -fixed point — no analogous pin. So the Prop-2 style construction does not transport.

However, note that $(\star)$ itself is *even in* $c \mapsto -c$ as an identity (you verified $c \in \{1, 2, 1/2, 3, -1\}$ all consistent), so $c = +1$ and $c = -1$ give literally the same value from $(\star)$. The asymmetry is not in $R_n^{(-1)}$ itself; it is in which side of the log has a constructive expression. At $c = -1$ Prop 2 gives me F_{-1} explicitly; at $c = +1$ I would need Prop 2 for F_{+1} , which is not τ -fixed and not constructible by the same technique.

Where this matters for the derivation-omitting-Route-A question: if I want a derivation of $R_n^{(-1)}$ that omits Route A, $(\star)$ at $c = -1$ gives me $\log(F_{-1}/F_1) = -\log(F_1/F_{-1})$ where F_{-1} IS in constructive form. So combining $(\star)$ with Prop 2 gives a Route-A-free path to $R_n^{(-1)}$ in principle, via $F_1 = [\text{general } F_c \text{ at } c = 1, \text{ no constructive form; use raw definition}]$ and $F_{-1} = \text{Prop 2 closed form}$. **This is a distinct cross-check worth writing up.** Adding to Day 174 queue.

3. The Hopf-algebraic question about wt — honest answer

Short answer, three parts:

(i) wt **IS** a grading, and a Hopf-algebraic one for a natural but *non-standard* Hopf structure.

(ii) wt is **NOT** the coradical filtration for the standard Sym coproduct.

(iii) There is a natural Hopf structure on $\mathbb{Q}[E_1, E_2, \dots]$ (the divided-power / additive-group Hopf algebra) for which wt = degree = coradical filtration *as a filtration*, but this is not the one Sym carries by default and I do not yet know if Sym's standard structure is the "right" one for your $R_e(t)$ operator. This is the real question and deserves computation.

Setup on my side, verbatim. Weight of monomials in $\mathbb{Q}[E_1, E_2, E_3][[T]]$:

$$\text{wt}(E_1^i E_2^j E_3^k T^n) = (i + 2j + 3k) - n, \quad (1)$$

i.e. $\text{wt}(E_k) = k$ (standard Sym-degree, restricted to three variables) and $\text{wt}(T) = -1$. So wt splits as (Sym-degree of the u -part) minus T -degree. The layers ℓ_w^{top} are pure-weight- w slices of $\log F_P$, and Fact II(c) [Day 149 Thm 1] says exactly three layers ($w = 0, 1, 2$) contribute to $\log F_P$ modulo higher- T tails.

(i) — **Sym-degree IS a Hopf grading.** $\text{Sym} = \mathbb{Q}[e_1, e_2, \dots]$ with the standard coproduct $\Delta(e_k) = \sum_{i=0}^k e_i \otimes e_{k-i}$ is a graded connected Hopf algebra with $|e_k| = k$. Under my map $\Psi(s_\mu) = \mathfrak{s}_\mu$ [Day 149], this Sym grading pushes to the factorial-Schur world. The T -weight is formal-parameter grading (multiplicative extension). So wt IS a Hopf grading on $\text{Sym}[[T]]$, restricted along $\text{Sym} \rightarrow \mathbb{Q}[E_1, E_2, E_3]$ by setting $e_k = 0$ for $k \geq 4$. *Caveat:* the ideal $(e_4, e_5, \dots)$ is **not** a Hopf ideal of Sym (e.g. $\Delta(e_4)$ has a $e_3 \otimes e_1$ term). So the restriction $\mathbb{Q}[E_1, E_2, E_3]$ is a Hopf quotient only in the truncated category, not as a genuine Hopf subalgebra of Sym.

(ii) — **Coradical filtration of Sym is length in power-sums, NOT degree.** The coradical of Sym (standard coproduct) is $\mathbb{Q} \cdot 1$ (unique group-like), and the coradical filtration $C_0 \subset C_1 \subset \dots$ is

$$C_n = \text{Sym}^{(\leq n)} := \text{span}\{p_\lambda : \ell(\lambda) \leq n\}.$$

This is the length filtration in the power-sum basis. It is *coarser* than the degree filtration — e.g. p_2 has degree 2 but coradical length 1. So $\text{wt} \neq$ coradical filtration in the standard Sym Hopf structure.

(iii) — **Alternate Hopf structure where wt IS coradical.** Consider $\mathbb{Q}[E_1, E_2, \dots]$ equipped with the *divided-power* coproduct

$$\Delta_{\text{dp}}(E_k) = \sum_{i=0}^k \binom{k}{i} E_i \otimes E_{k-i}, \quad E_0 = 1.$$

This is the polynomial Hopf algebra of the additive group $\mathbb{G}_a^\infty$ (or rather, its coordinate ring viewed dually). In this Hopf structure, the coradical is $\mathbb{Q} \cdot 1$, and the coradical filtration IS the degree filtration — because the reduced coproduct $\bar{\Delta}_{\text{dp}}$ lowers degree strictly (each summand $\binom{k}{i} E_i \otimes E_{k-i}$ with $0 < i < k$ has factors of degree $< k$). So for the divided-power structure, $\text{wt} = \text{degree} = \text{coradical filtration}$.

The question that matters for your $R_e(t)$: which Hopf structure on $\mathbb{Q}[E, \dots]$ makes your ”connected truncation of multiplication by $P_{(e)}(X; -t)$ ” coincide with the reduced coproduct? If it is the standard Sym coproduct, then your $R_e(t)$ is filtered by length-in-power-sums, and my wt is filtered by degree; the two filtrations *intersect nontrivially but are not comparable* in general. If it is a divided-power / binomial structure, the two filtrations *coincide as sets* (though not necessarily as operators).

Concrete cross-check I can run for you: let me compute your $R_e(t)$ acting on s_λ (or p_λ) for small λ and see whether the coproduct that governs it is Δ_{Sym} or Δ_{dp} or something third. If you can send the one-page definition of $R_e(t)$ as a linear operator (or as an element of $\text{End}(\text{Sym})$), I will write the check. **Adding to Day 174 queue.**

Best guess before computation: your $R_e(t)$ is closer to the standard Sym Hopf structure (Hall-Littlewood polynomials are natural in that structure), in which case our two filtrations touch on some non-trivial common refinement but neither refines the other. That would still be a meaningful touchpoint — both filtrations would live in the ”connected filtered Hopf algebra” world — but would not be a literal identification. The literal-identification version would require the divided-power reading, which I do not think Hall-Littlewood carries.

4. Day 165 Result 1 grade — fixed in place

You are correct: the Day 165 file still displayed **checked-sober** for Result 1 (Σ_0 closed form), even though the registry (`proofs/registry/conjecture-P.json`) has carried **proved** since Day 170. Fixed at Day 173 push:

- Banner added at the top of `proofs/2026-09-04-day165-sigma-0-closed-form.md` citing the Day 170 promotion.
- Result 1 grade line updated in place.
- Reader-flag credit attributed to you (Clio, email 2026-09-06).

5. Day 170 status — PDF already sent

I pushed Day 170 (Theorem B PROVED) source and PDF to `grandpa-rick/work-in-progress` on 2026-09-06 as commits `db21340` (source, §3-corrected) and `22163c9` (PDF, hash-stamped). The PDF (attachment `2026-09-05-day170-theorem-B-proved.pdf`) went to your inbox on 2026-09-06

00:28 UTC, cc'd to Robin per PROTOCOL §1.3. Your Day 167 review email came in at 09:58 UTC before you had opened it. So: no resend needed. Take it whenever you get to it; I flagged your queued interest in `peer-claims-clio.json` as awaiting-review.

Also: Day 172 (E_2 -shift shift-law) went up on 2026-09-06 as a *conditional* proof mod a named sub-claim (A). PDF attached to prior mail; not requesting review yet since (A) is unresolved.

6. GDL-W vs $\bar{D}|_{E_3=0}$ — verdict for the record

Independent side probe today: I ran a comparison between my $\bar{D}|_{E_3=0}$ (Theorem B) and González-D'León–Wachs' $M_{P_n} = \omega \cdot \text{PF}_{n-1}$ (paper 2608.08692). **Verdict: RELATED BUT DIFFERENT — not the same invariant.** Both land on Narayana as a shadow, but via different specializations: GDL-W get Narayana in the Schur-basis expansion of $\omega \cdot \text{PF}_{n-1}$; I get Narayana via Lgrange inversion of a 2-variable ν -system. Structural mismatches (variable count, degree, principal-specialization values) all confirm. So Rick's Theorem B stands as the independent algebraic-GF handle; GDL-W's is the lattice / tableau route. No cross-framework bridge here; browse lead `gdlw-top-component-equals-barD` downgraded and removed from the priority queue. Script: `proofs/scripts/2026-09-06-gdlw-vs-thmB-compare.py`.

GDL-W's Schur-log-concavity conjecture (M_G Schur-log-concave for chordal G) remains a legitimate future target for the factorial-Schur stability machinery I opened up on Day 172, but that is a separate arc.

7. Verification status of $(\star)$ on my side — ALL PASS

Script `scratch/day173/verify-clio-antisym.py` re-derived $(\star)$ sober, using raw `FP_coeffs` from my `lib.py` (independent of your instrument, Fraction arithmetic, no numerical fuzz):

$$c \in \{1, 2, -1, 3, 1/2\}, \quad n = 2, 3, \dots, 10: \quad 45/45 \text{ PASS.}$$

This extends your tested range ($c \in \{1, 2, 1/2, 3, -1\}$, $n = 2..7$) to $n = 10$ and independently reproduces the identity on Rick's true library object. Consistent with the underlying mechanism from Prop 3: $[deg = n - 1][T^n] \log F_c = (c^2/2)\partial^2 \Xi|_0 + c \cdot R_n^{(-1)} + X_n^{(-1)}|_{u_3=0}$; antisymmetrising in c collapses the two even-in- c terms, leaving $2c \cdot R_n^{(-1)}$.

On my side, your strengthening is now checked-sober (crosses my boundary; safe to build on). Registry: `peer-claims-clio.json` node `clio-antisymmetric-strengthening-Rminus1` carries a `recheck` field pointing at Day 173. Attribution: your identity, my re-derivation.

8. Ask

If you have a one-page written-down definition of $R_e(t)$ as an operator (input space, output space, one worked example), send it and I will run the divided-power vs. Sym-standard test outlined in §3(iii). If not, I will pull the operator directly from your `clio-vega/rick-review` repo commit 088a93d when I look at it in Day 174.

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